

w^+ w^+
 physical $\rightarrow$ $\mathcal{P}_i$
 $\in \mathcal{P} \in \mathcal{L}(\mathcal{P})$ w^- w^-
 (etc.)
 unitary gauge

$= \sum_{\text{all diag.}}^{w^+}$



(v)
 $G_{\mu\nu}(v, m)$

(v)

$$S_{\mu\nu}(k,m) = \text{diagram: } \nu \text{ --- } \bigcirc \text{ --- } \mu$$

$$= \text{diagram: } \nu \text{ --- } \checkmark \text{ --- } \mu + \text{diagram: } \nu \text{ --- } \bigcirc \text{ --- } \mu$$

+ a lot.

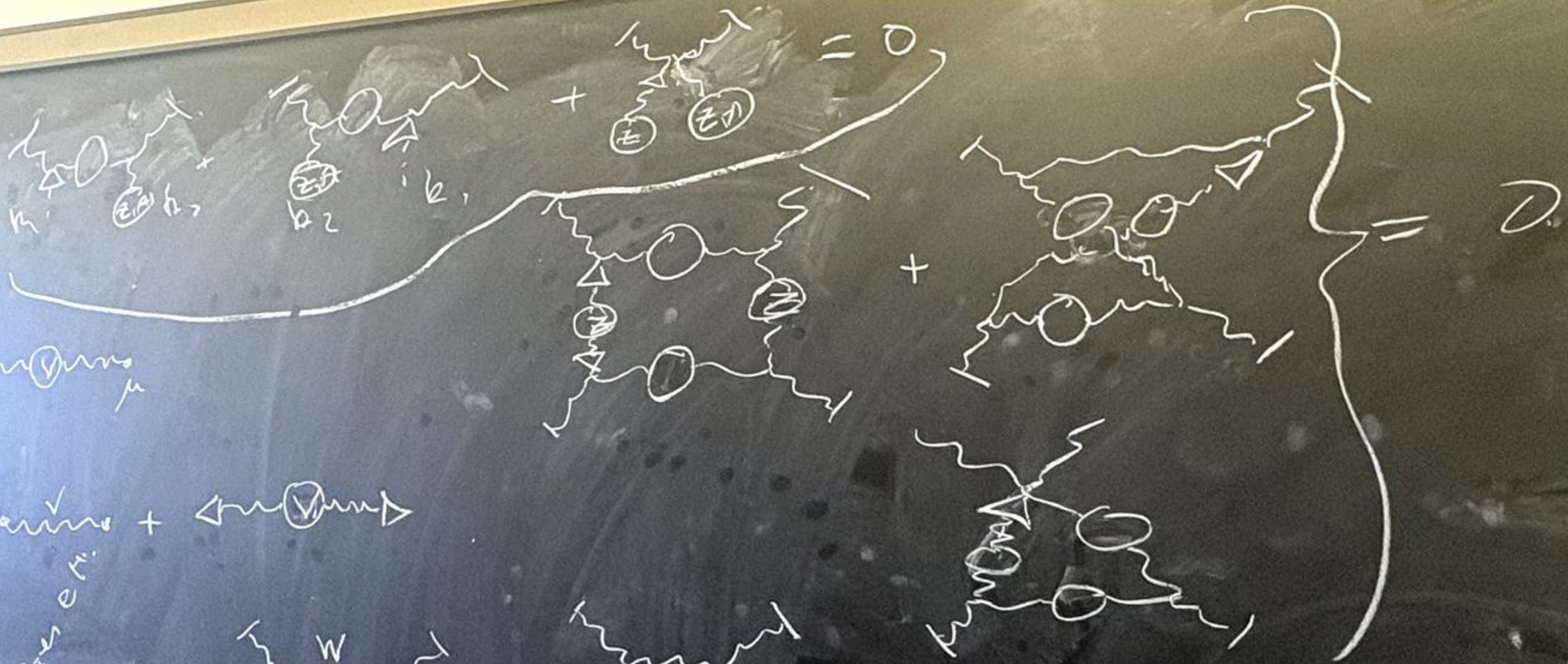
small
work
identities, etc.

(smaller
much
better UV
beh. (at most
log c))



verify "unitary
beh."

Also logⁿ
study beh.
in H.E.

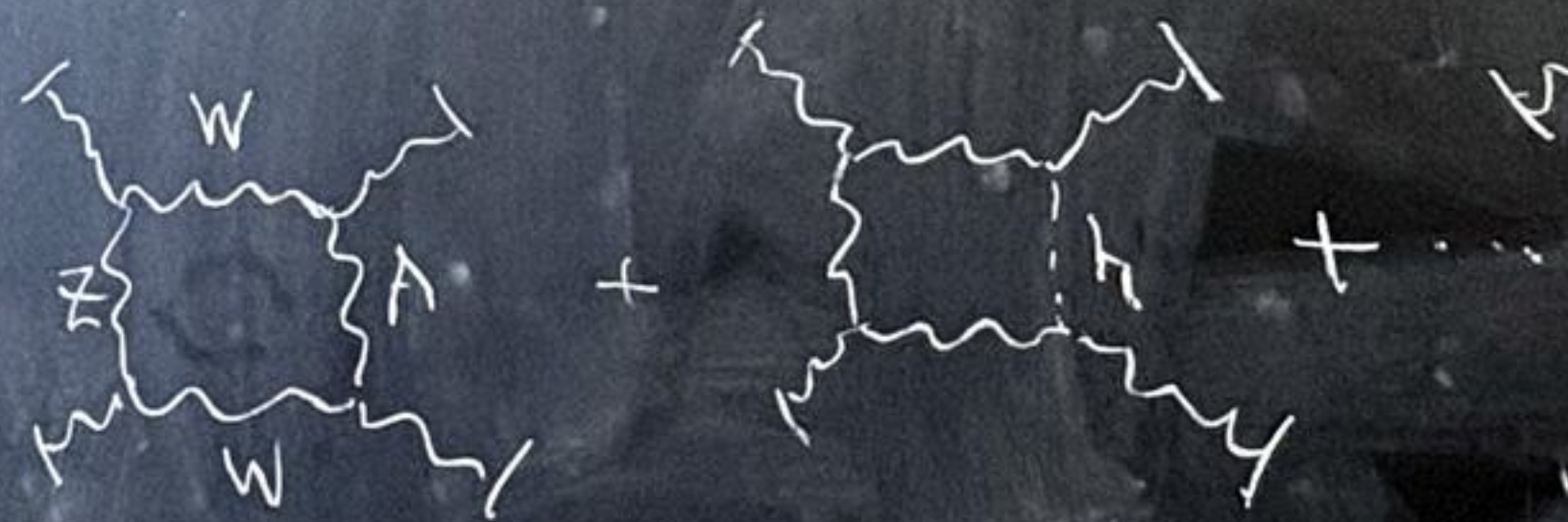


(VI)
 $G_{\mu\nu}(k,p) = \text{diagram}$

$= \text{diagram} + \text{diagram}$

small
 work
 using identities etc.

$\sum$



(smaller
 much
 better UV
 beh. (at most
 log))

verify "unitary
 beh."

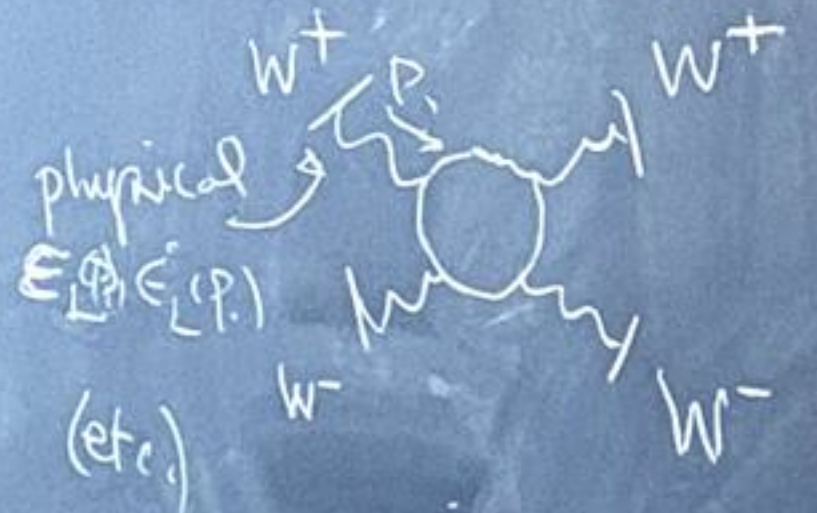
Also study log.
 study log.
 in H.E. beh.

can't forget



$$G_{\mu\nu}^{(V)}(k,m)$$

$$= \text{diagram with a circle containing a V}$$



unitary gauge

$$= \sum_{\text{all diag.}} w+$$

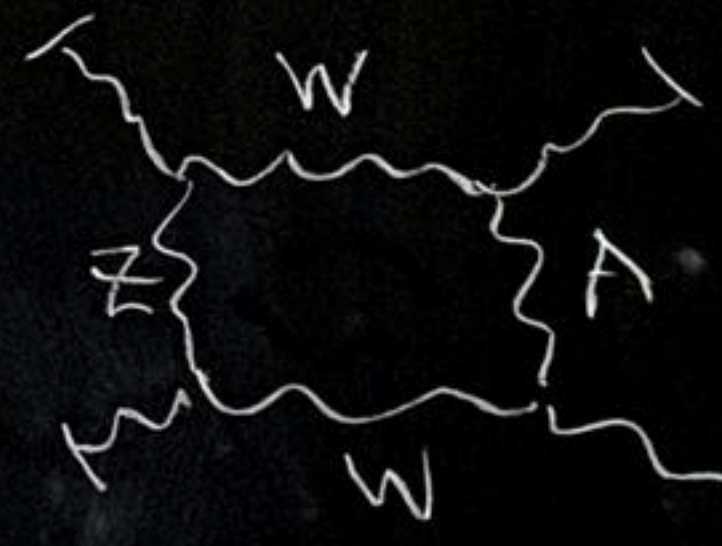


$$= \text{diagram with a circle containing a V} + \text{diagram with a circle containing a V and arrows}$$

all work using identities etc.

$$\sum$$

(smaller much better UV beh. (at most logic))



$$\begin{array}{c}
 \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} = 0
 \end{array}$$

Diagram 1: A fermion line with a loop containing a scalar and a fermion. Diagram 2: A fermion line with a loop containing a scalar and a fermion, with an external fermion line. Diagram 3: A fermion line with a loop containing a scalar and a fermion, with an external fermion line.

$\Gamma(m)$

$$= \text{Diagram 4}$$

Diagram 4: A fermion line with a loop containing a scalar and a fermion.

$$\begin{array}{c}
 \text{Diagram 5} + \text{Diagram 6} + \text{Diagram 7}
 \end{array}$$

Diagram 5: A fermion line with a loop containing a scalar and a fermion. Diagram 6: A fermion line with a loop containing a scalar and a fermion. Diagram 7: A fermion line with a loop containing a scalar and a fermion.

$$= \text{Diagram 8} + \text{Diagram 9}$$

Diagram 8: A fermion line with a loop containing a scalar and a fermion. Diagram 9: A fermion line with a loop containing a scalar and a fermion.

more work using identities etc.

Σ

(smaller much better UV beh. (at most log c))

$$\begin{array}{c}
 \text{Diagram 10} + \text{Diagram 11} + \dots
 \end{array}$$

Diagram 10: A fermion line with a loop containing a scalar and a fermion. Diagram 11: A fermion line with a loop containing a scalar and a fermion.

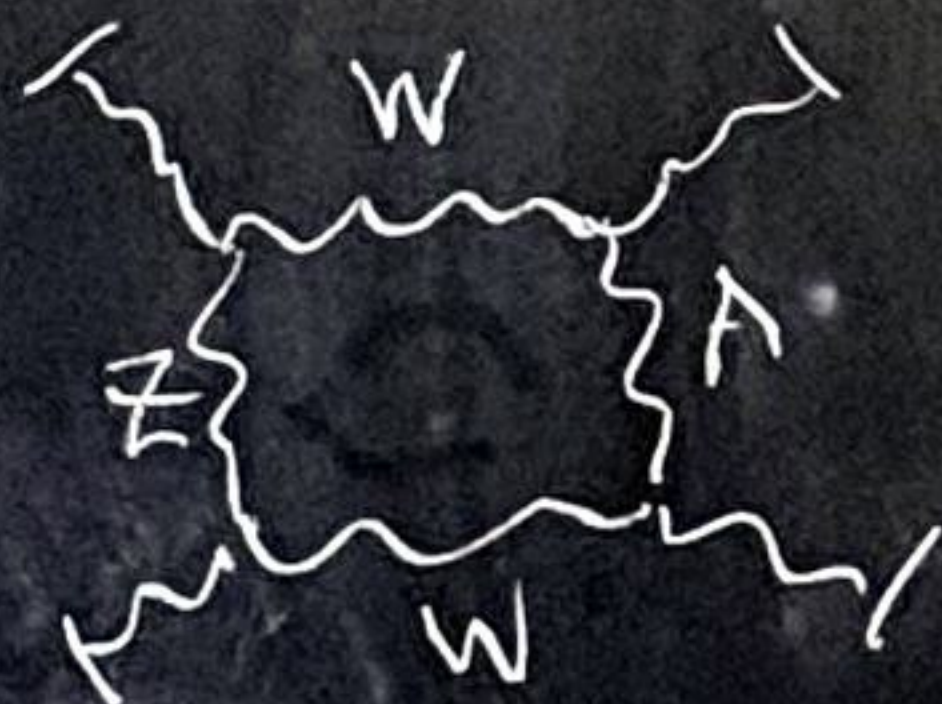
Verify "unitary beh."

Also log study in H.E. beh.

more
wavy identities etc.

Σ

(smaller
much
better UV
beh. (at most
logic))



Verify "unitary
beh."

Also
sto
in